Aman Preet Singh

Distributed Systems & Go Backend Engineer

Roll No: 22CS110 | Department: Computer Science & Engineering | CGPA: 9.20 / 10.0

Email: aman.singh@college.edu | Phone: +91 95555 44556 | LinkedIn: linkedin.com/in/aman-preet-systems

PROFESSIONAL SUMMARY

High-concurrency systems engineer specializing in Golang, gRPC protocols, microservice mesh architecture, distributed consensus algorithms, and PostgreSQL database performance.

CORE TECHNICAL SKILLS

- **Languages:** Go (Golang), C++, SQL, Bash, Python
- **Distributed Systems:** gRPC, Protocol Buffers, Raft Consensus, Redis Pub/Sub, Kafka
- **Databases & Cache:** PostgreSQL, Redis, ClickHouse, SQLite
- **DevOps & Mesh:** Kubernetes, Docker, Envoy, OpenTelemetry, Grafana, Linux

WORK EXPERIENCE & INTERNSHIPS

Systems Engineering Intern | *CloudPulse Networks* May 2024 - Aug 2024

- Engineered Go microservices processing 45,000 gRPC requests/sec with p99 latency under 4 milliseconds.
- Deployed distributed tracing using OpenTelemetry & Jaeger to monitor call graphs across 15 microservices.

FEATURED TECHNICAL PROJECTS

Distributed Key-Value Store with Raft Consensus *(Go, gRPC, Protobuf, Docker)*

- Developed fault-tolerant replicated key-value storage engine implementing Raft consensus and leader heartbeats.
- Supported snapshotting and log compaction to minimize disk usage and state recovery times.

High-Throughput Flash Sale Inventory Engine *(Golang, PostgreSQL, Redis, Kubernetes)*

- Engineered atomic reservation engine preventing overselling with distributed Redis locks and optimistic concurrency.

EDUCATION & ACADEMIC BACKGROUND

Bachelor of Technology in Computer Science & Engineering • Thapar Institute of Engineering & Technology 2022 - 2026

CGPA: 9.20 / 10.0 | Relevant Coursework: Distributed Systems, Operating Systems, Concurrent Programming

CERTIFICATIONS & HONORS

- Certified Kubernetes Application Developer (CKAD)
- Google Cloud Certified Associate Cloud Engineer